Study Report

AUTODESK

Analyzed File	CubeSat Structure - Main Body
Version	Autodesk Fusion (2606.1.36)
Creation Date	2026-02-27, 18:59:09
Author	Kumar Utkarsh

Report Properties

Title	Studies
Author	Kumar Utkarsh

Simulation Model 1

Study 3 - Modal Frequencies

Study Properties

Study Type	Modal Frequencies
Last Modification Date	2026-02-27, 18:53:53

Settings

General

Contact Tolerance	0.10 mm
Number of Modes	8
Frequency Range [Hz]	No
Compute Preloaded Modes	No

Mesh

Average Element Size (% of model size)	
Solids	8
Scale Mesh Size Per Part	No
Average Element Size (absolute value)	-
Element Order	Parabolic
Create Curved Mesh Elements	Yes
Max. Turn Angle on Curves (Deg.)	60
Max. Adjacent Mesh Size Ratio	1.5
Max. Aspect Ratio	10
Minimum Element Size (% of average size)	20

Adaptive Mesh Refinement

Number of Refinement Steps	0
Results Convergence Tolerance (%)	20
Frequency Mode	1

Materials

Component	Material	Safety Factor
Body1	PLA	Yield Strength

PLA

Density	1.240E-06 kg / mm^3
Young's Modulus	2800.00 MPa
Poisson's Ratio	0.36
Yield Strength	39.995 MPa
Ultimate Tensile Strength	49.599 MPa
Thermal Conductivity	1.600E-04 W / (mm K)
Thermal Expansion Coefficient	7.020E-05 / K
Specific Heat	1795.00 J / (kg K)

Contacts

Mesh

Type	Nodes	Elements
Solids	37419	19605

Load Case1

Results

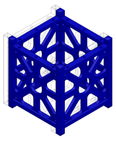
Result Summary

Frequency	Participation X	Participation Y	Participation Z
Mode 1: 0.00 Hz	100	0	0
Mode 2: 0.00 Hz	0	100	0
Mode 3: 0.00 Hz	0	0	100
Mode 4: 0.00 Hz	0	0	0
Mode 5: 0.00 Hz	0	0	0
Mode 6: 0.00 Hz	0	0	0
Mode 7: 259.493 Hz	0	0	0
Mode 8: 280.922 Hz	0	0	0

Total Modal Displacement

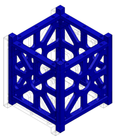
Mode 1: 0.00 Hz Total Modal Displacement

1.00 1.00



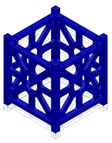
Mode 2: 0.00 Hz Total Modal Displacement

1.00 1.00



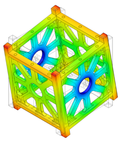
Mode 3: 0.00 Hz Total Modal Displacement

1.00 1.00



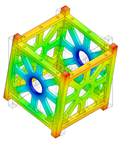
Mode 4: 0.00 Hz Total Modal Displacement

0.16 1.00



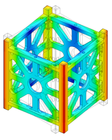
Mode 5: 0.00 Hz Total Modal Displacement

0.16 1.00



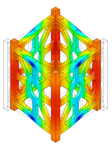
Mode 6: 0.00 Hz Total Modal Displacement

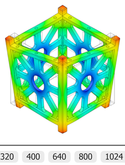
0.593 1.00



Mode 7: 259.493 Hz Total Modal Displacement

0.588 1.00



Mode 8: 280.922 Hz Total Modal Displacement

0.11 1.00